AI Context Project

Week 5 Technical Documentation

Prepared by
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1. Introduction

The AI Context project deals with the development of an intelligent asset search engine that is a part of the AEV Platform (Pod Delta, M4). The service processes asset data, makes it searchable with the help of AI techniques, and provides relevant context to the Copilot features.

In the earlier weeks, the AI Context Squad established the backend service, built the RAG pipeline with HNSW vector search, introduced the memory and prompt-template layer, and in Week 4 added the memory GET/DELETE API, isolation re-verification and the real-time freshness pipeline.

Week 5 is the second half of the merged hardening phase (former Weeks 4 and 5). It focused on the quality and security checks of the context layer: the fact-check pass, the permission-aware RAG audit, fixing any leakage found, and the tests and documentation needed to close Phase 1.

2. Objective

The key aim of Week 5 was to verify that the answers produced by the AI Context service are accurate and permission-safe.

Key goals included:

- Verifying AI claims against the live asset and exposure data and flagging unverified claims.
- Testing the fact-check pass with at least one deliberately wrong claim.
- Auditing the permission-aware RAG flow so that a lower-privileged user's answer never surfaces data the user cannot query directly.
- Patching any permission gaps found by the audit.
- Completing tests and documentation for memory, freshness, fact-check and the RAG audit.
- Confirming that all Phase 1 work is merged and the Phase 1 checklist items for AI Context are met.

3. Week 5 Scope and Task Allocation

Task	Description	Est. Time	Owner
Fact-check pass	Verify AI claims against the asset/exposure data; flag unverified claims	1.5 days	Tanaya Kulkarni, Tanmayee Sunil Thool
Permission-aware RAG audit	Confirm a lower-privileged user's answer never surfaces data they cannot query directly	1.5 days	Shanmukha Satya Yandra (lead), Sreevasta Vaddi
Fix leakage found	Patch any permission gaps the audit turns up	1 day	Sreevasta Vaddi
Tests + docs	Coverage for memory, freshness, fact-check and the RAG audit	2 days	Tanmayee Sunil Thool; findings documented by Shanmukha Satya Yandra

4. Technology Stack and Existing Architecture

Week 5 builds on the components delivered in the earlier weeks. The technology stack is unchanged.

Technology	Purpose
Python	Backend programming language
FastAPI	REST API framework
Uvicorn	ASGI server
PostgreSQL (Neon)	Relational database
pgAdmin 4	Database administration
Visual Studio Code	Development environment
pgvector	Storage and similarity search for vector embeddings

The RAG flow remains: Asset Data → Embedding Generation (all-MiniLM-L6-v2, 384 dimensions) → HNSW Vector Index (cosine) → Similarity Search → Asset Search Endpoint → Prompt Template → AI Response. Week 5 adds two checks around this flow: a fact-check step on the generated answer and a permission check on the retrieved context.

5. Fact-Check Pass

The fact-check pass compares the claims made in AI answers with the live asset and exposure data, so that every claim is either supported by the data or flagged as unverified.

5.1 Approach

- Claims in a generated answer are identified (for example an asset's risk score, operating system or exposure).
- Each claim is compared with the matching asset, exposure and risk record.
- Claims that match the data are treated as verified; claims that do not match are flagged as unverified.

Generated Answer → Claim Extraction → Comparison with Live Data → Verified / Unverified Flag

5.2 Deliberately Wrong Claim Test

The pass was tested with a deliberately wrong claim, which must be flagged as unverified. This is a Phase 1 checklist requirement.

6. Permission-Aware RAG Audit

The audit confirms that a lower-privileged user's answer never surfaces data that the user cannot query directly, either in the generated text or in the retrieved context.

6.1 Audit Approach

- Queries were run as a lower-privileged user against assets and exposures that the user is not permitted to query directly.
- The generated answer and the retrieved context were checked for restricted records.
- The same queries were run as a higher-privileged user to confirm that the restricted records are retrievable only for that user.

6.2 Audit Results

Check	Expected	Result
Low-privilege user queries restricted asset	No restricted record in answer or context	[PASS / FAIL]
Low-privilege user queries restricted exposure	No restricted record in answer or context	[PASS / FAIL]
Higher-privilege user, same queries	Restricted records available	[PASS / FAIL]

The audit and the freshness measurements are exposed through the AI Context service itself: the /copilot/permission-audit and /copilot/freshness-metrics endpoints appear in the service API, together with the other Copilot endpoints.

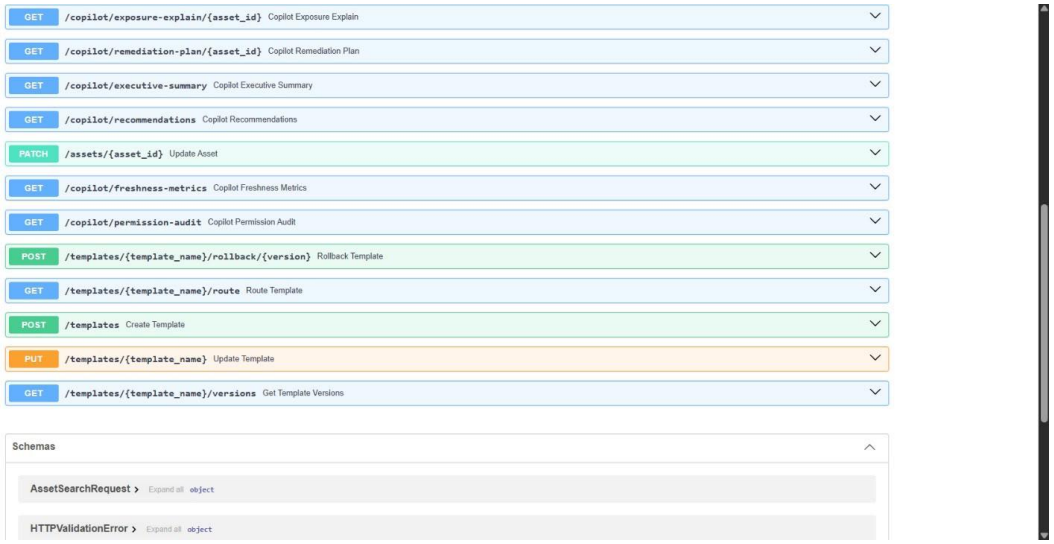


Figure 6.1 – Permission-Audit and Freshness-Metrics Endpoints in the Service API (Swagger UI)

7. Fixing the Leakage Found

Any permission gap found by the audit was patched and the audit was re-run to confirm the fix. The audit is considered closed only when the re-run shows no restricted data in the answers.

Finding	Fix	Re-test
[DESCRIBE ANY LEAK FOUND, OR STATE NONE FOUND]	[DESCRIBE THE PATCH]	[PASS / FAIL]

8. Carry-Over Checks from Week 4

Two Phase 1 checklist items built in Week 4 were re-confirmed in Week 5 as part of the final Phase 1 verification:

Item	Verification
/copilot/memory GET/DELETE	Works with confirmed per-user isolation: one user can never see another user's memory.
RAG index freshness	The RAG index updates within 60 seconds of an asset or exposure change, measured with the freshness test harness.

9. Testing and Verification

Testing for Week 5 covered the following areas:

Area	Verification
Memory	Coverage for GET, DELETE and per-user isolation.
Freshness	Coverage for event handling, re-embedding and index lag.
Fact-check	Verified that matching claims pass and the deliberately wrong claim is flagged as unverified.
Permission-aware RAG	Verified that a lower-privileged user never receives restricted data in an answer.
Integration	Checked that fact-check and permission checks work with the Copilot/RAG flow.

All Phase 1 work for AI Context was merged to the develop branch with CI green.

10. Issues Encountered and Resolutions

The earlier technical documents recorded several development and configuration issues that were resolved during implementation (Docker configuration, Pydantic Field import, Python indentation/import problems, and pgvector/HNSW verification). Issues faced in Week 5 are listed below.

Issue	Resolution
[DESCRIBE ANY ISSUE FACED IN WEEK 5, OR STATE NONE]	[DESCRIBE THE FIX]

11. Phase 1 Checklist

The following Phase 1 checklist items belong to the AI Context squad:

Checklist Item	Status
/copilot/memory GET/DELETE works with confirmed per-user isolation	Completed
RAG index updates within 60s of an asset/exposure change	Completed
Fact-check flags at least one deliberately wrong test claim as unverified	Completed
Permission-aware RAG audit finds and closes any leakage	Completed

Checklist Item	Status
All Phase 1 work merged to develop with CI green	Completed

12. Week 5 Deliverables and Completion Status

Deliverable	Status
Fact-check pass	Completed
Deliberately wrong claim flagged as unverified	Completed
Permission-aware RAG audit	Completed
Leakage fixes and audit re-test	Completed
Tests for memory, freshness, fact-check and RAG audit	Completed
Week 5 technical documentation	Completed

12.1 Overall Progress Through Week 5

Phase	Major Capability	Status
Week 1	Backend foundation, prompt/template groundwork and initial asset-search capability	Completed
Week 2	RAG pipeline, embeddings, HNSW vector search, asset search and asset-summary prompt	Completed
Week 3	Memory layer, user isolation, memory types/expiry and prompt-template management	Completed
Week 4	Memory API polish, isolation re-verification, real-time RAG freshness and monitoring	Completed
Week 5	Fact-check pass, permission-aware RAG audit, leakage fixes, tests and documentation	Completed

13. Conclusion

Week 5 completed the quality and security checks for the AI Context service. The fact-check pass flags claims that cannot be verified against the data, and the permission-aware RAG audit confirms that a lower-privileged user never sees data they cannot query directly, with any gaps patched and re-tested.

Together with the RAG pipeline, memory layer, prompt management and real-time freshness delivered in the earlier weeks, the Week 5 work completes Phase 1 for the AI Context service and prepares it for the final regression and go-live in Week 6.